

Practice Problems: Counting

1 Review Problems

Note 1. Your quiz will have python review problems, but they will be in a different format. Instead of being presented as "raw" python, they will be embedded with the shell.

Problem 1. Write the output of the final command in the following terminal session. If the command has no output, then leave the problem blank.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
xs = [1, 2, 3]
def foo(y):
    xs.append(y)
    return y
foo(5)
print('sum(xs)=', sum(xs))
EOF
$ python3 foo.py
```

2 Nested for loops

Note 2. For the rest of these problems, the goal is to learn shortcuts to counting the number of times certain lines will get run. For nested for loops, we multiply the number of iterations of the loop; for sequential loops, we add.

Problem 2. Write the output of the final command in the following terminal session. If the command has no output, then leave the problem blank.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ N=8
$ cat > foo.py <<EOF
count = 0
for i in range($N):
    count += 1
print('count=', count)
EOF
$ python3 foo.py
```

Problem 3. Write the output of the final command in the following terminal session. If the command has no output, then leave the problem blank.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ N=8
$ cat > foo.py <<EOF
count = 0
for i in range($N):
    for j in range($N):
        count += 1
print('count=', count)
EOF
$ python3 foo.py
```

Problem 4. Write the output of the final command in the following terminal session. If the command has no output, then leave the problem blank.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ N=8
$ cat > foo.py <<EOF
count = 0
for i in range($N):
    for j in range($N):
        for k in range($N):
            count += 1
print('count=', count)
EOF
$ python3 foo.py
```

3 Sequential for loops

Problem 5. Write the output of the final command in the following terminal session. If the command has no output, then leave the problem blank.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ N=8
$ cat > foo.py <<EOF
count = 0
for i in range($N):
    count += 1
for i in range($N):
    count += 1
print('count=', count)
EOF
$ python3 foo.py
```

Problem 6. Write the output of the final command in the following terminal session. If the command has no output, then leave the problem blank.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ N=8
$ cat > foo.py <<EOF
count = 0
for i in range($N):
    count += 1
for i in range($N):
    count += 1
for i in range($N):
    count += 1
print('count=', count)
EOF
$ python3 foo.py
```

4 Combinations of sequential and nested loops

Problem 7. Write the output of the final command in the following terminal session. If the command has no output, then leave the problem blank.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ N=8
$ cat > foo.py <<EOF
count = 0
for i in range($N):
    for j in range($N):
        count += 1
for i in range($N):
    count += 1
    count += 1
count += 1
print('count=', count)
EOF
$ python3 foo.py
```

Problem 8. Write the output of the final command in the following terminal session. If the command has no output, then leave the problem blank.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ N=8
$ cat > foo.py <<EOF
count = 0
for i in range($N):
    count += 1
    for j in range($N):
        count += 1
for i in range($N):
    count += 1
print('count=', count)
EOF
$ python3 foo.py
```

Problem 9. Write the output of the final command in the following terminal session. If the command has no output, then leave the problem blank.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ N=8
$ cat > foo.py <<EOF
count = 0
for i in range($N):
    count += 1
for i in range($N):
    count += 1
    for j in range($N):
        count += 1
        count += 1
    for j in range($N):
        count += 1
        for k in range($N):
            count += 1
for i in range($N):
    count += 1
print('count=', count)
EOF
$ python3 foo.py
```

5 “Simple” Expressions in Multiple Variables

Problem 10. Write the output of the final command in the following terminal session. If the command has no output, then leave the problem blank.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ N=8
$ M=4
$ cat > foo.py <<EOF
count = 0
for i in range($N):
    for j in range($M):
        count += 1
print('count=', count)
EOF
$ python3 foo.py
```

Problem 11. Write the output of the final command in the following terminal session. If the command has no output, then leave the problem blank.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ N=8
$ M=4
$ cat > foo.py <<EOF
count = 0
for i in range($N):
    count += 1
for j in range($M):
    count += 1
print('count=', count)
EOF
$ python3 foo.py
```

Problem 12. Write the output of the final command in the following terminal session. If the command has no output, then leave the problem blank.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ N=8
$ M=4
$ cat > foo.py <<EOF
count = 0
for i in range($N):
    count += 1
    for j in range($M):
        count += 1
        for j in range($N):
            count += 1
for i in range($M):
    count += 1
    for j in range($M):
        count += 1
        count += 1
        count += 1
print('count=', count)
EOF
$ python3 foo.py
```

Problem 13. Write the output of the final command in the following terminal session. If the command has no output, then leave the problem blank.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ N=8
$ M=4
$ O=2
$ cat > foo.py <<EOF
count = 0
for i in range($N):
    count += 1
    for j in range($M):
        count += 1
        for j in range($O):
            count += 1
for i in range($M):
    count += 1
    for j in range($O):
        count += 1
        count += 1
        count += 1
print('count=', count)
EOF
$ python3 foo.py
```

6 “Complex” Expressions in Multiple Variables

Problem 14. Write the output of the final command in the following terminal session. If the command has no output, then leave the problem blank.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ N=8
$ M=4
$ O=2
$ cat > foo.py <<EOF
count = 0
for i in range($N * $M):
    for j in range($M * $O):
        count += 1
print('count=', count)
EOF
$ python3 foo.py
```

Problem 15. Write the output of the final command in the following terminal session. If the command has no output, then leave the problem blank.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ N=8
$ M=4
$ O=2
$ cat > foo.py <<EOF
count = 0
for i in range($N + $M):
    for j in range($M * $O):
        count += 1
print('count=', count)
EOF
$ python3 foo.py
```

Problem 16. Write the output of the final command in the following terminal session. If the command has no output, then leave the problem blank.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ N=8
$ M=4
$ O=2
$ cat > foo.py <<EOF
count = 0
for i in range($N + $M):
    count += 1
for i in range($M * $O):
    count += 1
print('count=', count)
EOF
$ python3 foo.py
```

Problem 17. Write the output of the final command in the following terminal session. If the command has no output, then leave the problem blank.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ N=8
$ M=4
$ O=2
$ cat > foo.py <<EOF
count = 0
for i in range($N + $M):
    count += 1
    for j in range($N ** $O):
        count += 1
for i in range($M * $O):
    count += 1
print('count=', count)
EOF
$ python3 foo.py
```

Problem 18. Write the output of the final command in the following terminal session. If the command has no output, then leave the problem blank.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ N=8
$ M=4
$ O=2
$ cat > foo.py <<EOF
count = 0
for i in range($N + $M):
    count += 1
for i in range($N ** $O):
    count += 1
    for j in range($M + $O):
        count += 1
print('count=', count)
EOF
$ python3 foo.py
```

Note 3. The only math operations I will use in the real quiz are addition, multiplication, and exponentiation.